Mechanistic Hypothesis Search — Primary Hyperoxaluria Type 3

Cytosolic 4-Hydroxy-2-Oxoglutarate Cleavage Model

(`cytosolic_hog_cleavage`)

Disease: Primary Hyperoxaluria Type 3 (PH3) | **Category:** Metabolic Disorder **Hypothesis status in KB:** CANONICAL **Search date:** 2026-10-15 | **Iterations completed:** 5 | **Papers reviewed:** 24 | **Findings recorded:** 7

Executive Judgment

Verdict: PARTIALLY SUPPORTED but UNRESOLVED.

The cytosolic HOG-cleavage model is the field's default explanatory framework for why patients lose a mitochondrial enzyme upstream of glyoxylate yet overproduce oxalate. Its *upstream* half is firmly established: HOGA1 (DHDPSL) is a strictly mitochondrial enzyme whose PH3 variants are loss-of-function, and the hydroxyproline-pathway metabolites 4-hydroxy-2-oxoglutarate (HOG), 4-hydroxyglutamate, and 2,4-dihydroxyglutarate (DHG) accumulate in PH3 patient urine, serum, and liver. The hypothesis also rests on one piece of *direct* biochemical support — the detection of a cytosolic, non-HOGA1 HOG-aldolase activity in human hepatocyte cell lines (PMID:22729392).

However, the hypothesis's **defining causal step is unproven**. The responsible cytosolic aldolase has never been molecularly identified, and the one explicitly tested candidate — N-acetylneuraminate aldolase (NAL) — was nearly inactive against HOG (PMID:22771891). The HOG→glyoxylate→oxalate flux that the model requires has never been directly measured. The strongest counter-evidence comes from the Hoga1-knockout mouse, which accumulates HOG but does **not** become hyperoxaluric, and in which residual (non-HOGA) HOG→glyoxylate aldolase activity localizes to **liver mitochondria, not cytosol**, while the cytosolic compartment instead favors reductive diversion of HOG to DHG (PMID:26428388, PMID:39040543,

PMID:40578401). Finally, hepatic-LDH RNAi (nedosiran) fails to correct oxalate in PH3, implying a multi-organ oxalate source that the hepatocyte-centric terminal step of this model does not capture (PMID:39582099).

Two **competing, more parsimonious mechanisms** explain the same disease features without requiring an unidentified cytosolic aldolase: (1) HOG directly inhibits glyoxylate reductase (GR, the PH2 enzyme), impairing glyoxylate detoxification (PMID:22771891); and (2) HOGA1 loss reduces pyruvate availability and Krebs-cycle flux, indirectly raising oxalate (PMID:31696211). **Recommendation for curation:** retain the model as the default narrative but downgrade its mechanistic certainty — flag the cytosolic-aldolase edge as *speculative/unconfirmed* and register the two competing hypotheses as tracked alternatives with an explicit knowledge gap on direct flux measurement and enzyme identity.

Key Findings

F001 — Direct in vitro support: cytosolic non-HOGA1 HOG aldolase activity in human hepatocytes

The single most direct piece of evidence for the seed hypothesis comes from Belostotsky, Pitt & Frishberg (*Primary hyperoxaluria type III—a model for studying perturbations in glyoxylate metabolism*, PMID:22729392). Using GC-MS on PH3 patient urine, they documented significant elevations of HOG, its precursor 4-hydroxyglutamate, and its derivative 2,4-dihydroxyglutarate relative to mutation carriers and controls. Critically, in human hepatocyte cell lines they *detected a cytosolic 4-hydroxy-2-oxoglutarate aldolase activity* that was not attributable to HOGA1. The authors articulate the model's central route verbatim: HOG undergoes "its transport into the cytosol where it is converted to glyoxylate by a different cytosolic aldolase." This is the biochemical observation on which the entire cytosolic cleavage hypothesis is built.

Caveat: the finding is qualitative — an activity was detected, but the enzyme was not purified or identified, and no flux from HOG through glyoxylate to oxalate was quantified. The observation establishes *possibility*, not *operation in vivo*.

F002 — Qualifier: in Hoga1-KO mouse, residual HOG→glyoxylate activity is mitochondrial; cytosol favors reductive diversion to DHG

Li et al. (*4-hydroxy-2-oxoglutarate metabolism in a mouse model of Primary Hyperoxaluria Type 3*, PMID:39040543, *Biochem Biophys Rep* 2024) directly interrogated the compartmentalization the model depends on. In Hoga1-knockout mouse tissues, residual HOG→glyoxylate aldolase activity "was highest in liver mitoplasts, with no activity present in brain tissue lysates" — i.e., the non-HOGA aldolase activity is **mitochondrial, not cytosolic**, the opposite of what the seed model predicts. Moreover, "the metabolism of HOG to DHG was highest in liver and kidney cortical tissue, enhanced in the cytosolic compartment of the liver, and preferred NADPH as a cofactor." In other words, the murine cytosol preferentially *reduces* HOG to a dead-end (DHG) rather than *cleaving* it to glyoxylate. This directly qualifies the seed hypothesis at the level of its defining subcellular logic — at least in mouse.

F003 — Scope limitation: HOGA1-deficient mice do not recapitulate hyperoxaluria; human penetrance is low

The model must explain oxalate overproduction, yet the best in vivo model fails to overproduce oxalate. Li et al. (*Hydroxyproline metabolism in a mouse model of Primary Hyperoxaluria Type 3*, PMID:26428388, 2015) reported that Hoga1-null mice accumulate HOG, 4-OH-glutamate, and DHG but show **no increased oxalate excretion**, noting this "may be consistent with the recent recognition that only a small fraction of the individuals with the genotype for HOGA deficiency develop PH." Boffa/Brunetti-Pierri et al. (PMID:40578401, 2025) tested whether the murine non-phenotype was an artifact of species differences in AGT compartmentalization: even Hoga1-/-/Agxt-/- mice engineered to express AGT in peroxisomes only (the human-like configuration) "did not show increased urinary oxalate, suggesting that AGT expression in both cellular compartments is not an explanation for the lack of hyperoxaluria in Hoga1-/- mice." This both documents incomplete penetrance in humans and removes one candidate explanation for the mouse–human discrepancy, leaving the in vivo validity of the compartmentalization model unconfirmed.

F004 — Competing mechanism: reduced pyruvate / Krebs-cycle flux

Huang et al. (*Regulation of human 4-hydroxy-2-oxoglutarate aldolase by pyruvate and α-ketoglutarate*, PMID:31696211, 2019) characterized recombinant human HOGA1 and found it possesses dual aldolase **and** oxaloacetate-decarboxylase activities sharing the same catalytic machinery, both with micromolar substrate affinities likely operative in vivo, with pyruvate acting as a competitive inhibitor. They propose an alternative to cytosolic cleavage entirely:

"loss of HOGA1 function could increase oxalate production in PH3 by decreasing pyruvate availability and metabolic flux through the Krebs cycle." This is a *parallel, more parsimonious* mechanism that requires no unidentified cytosolic enzyme.

F005 — Competing mechanism + candidate-aldolase falsification: HOG inhibits glyoxylate reductase; NAL fails to cleave HOG

Riedel et al. (*4-Hydroxy-2-oxoglutarate aldolase inactivity in primary hyperoxaluria type 3 and glyoxylate reductase inhibition*, PMID:22771891, 2012) delivered three results that both anchor and challenge the model. First, they confirmed loss-of-function: all nine PH3 HOGA variants expressed in E. coli/CHO were unstable, aggregated, and retained no measurable activity. Second, they confirmed the upstream substrate accumulation on which both the seed and competing models depend: "a buildup of HOG was confirmed in the urine, sera and liver samples from PH3 patients." Third — and most consequentially — they tested two mechanistic candidates. N-acetylneuraminate aldolase, a plausible cytosolic HOG-cleaving aldolase, "showed minimal activity towards HOG," falsifying the leading named candidate for the seed's missing enzyme. And glyoxylate reductase "was inhibited by HOG but not by 2-hydroxyglutarate or 2-oxoglutarate," furnishing a competing mechanism in which accumulating HOG impairs glyoxylate detoxification directly.

F006 — Mechanistic history: PH3 hypothesis evolved from gain-of-function to loss-of-function

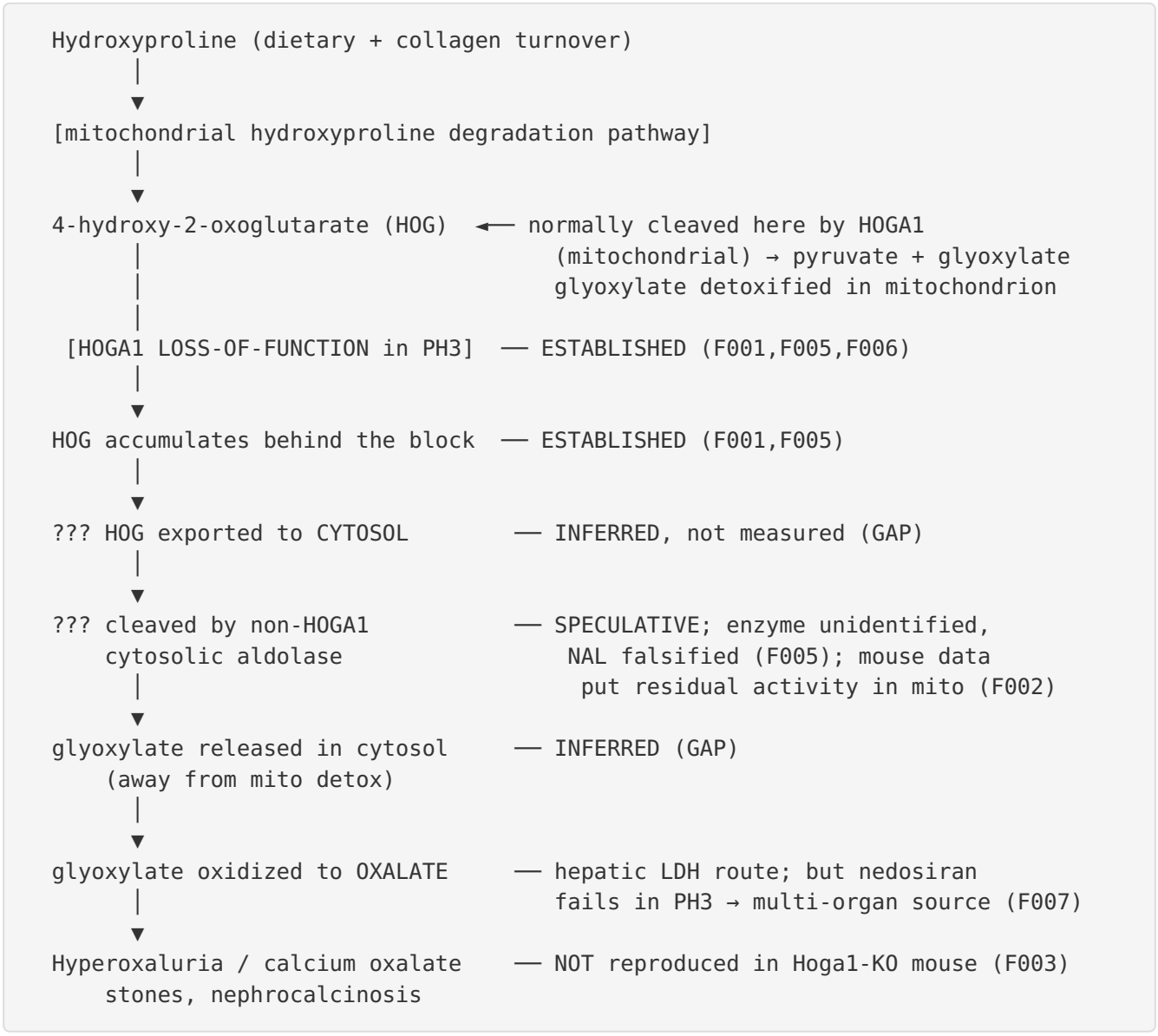
The downstream mechanism of PH3 has been unsettled from the start. Belostotsky et al. (*Mutations in DHDPSL are responsible for primary hyperoxaluria type III*, PMID:20797690, 2010, *Am J Hum Genet*) identified biallelic DHDPSL (HOGA1) mutations as the cause of PH3, establishing that "DHDPSL is the gene encoding 4-hydroxy-2-oxoglutarate aldolase, catalyzing the final step in the metabolic pathway of hydroxyproline." Monico et al. (PMID:21896830, 2011) initially "hypothesized to cause a gain of mitochondrial 4-hydroxy-2-oxoglutarate aldolase activity, resulting in excess oxalate," before revising to a loss-of-function model after mutation screening and RT-PCR. They also noted PH3's distinctive clinical signature: lower urine oxalate and higher urine calcium/uric acid than PH1/PH2, with generally preserved eGFR (mean 116 mL/min/1.73m^2 at 7.2 yr). The historical reversal itself illustrates that the paradox at the model's core — upstream block, downstream overproduction — has never had a settled resolution.

F007 — Qualifier: hepatic LDH inhibition (nedosiran) fails in PH3, implying a multi-organ oxalate source

Wanders et al. (*Human glyoxylate metabolism revisited: New insights pointing to multi-organ involvement*, PMID:39582099, *J Inherit Metab Dis* 2025, review-level synthesis) report that the hepatic-LDH-targeting RNAi nedosiran "effectively reduced urinary oxalate excretion in PH1 patients but not in PH2 or PH3 patients," and conclude from a thorough literature study that "glyoxylate/oxalate metabolism is not confined to the liver but instead involves multiple different organs." If the terminal oxalate-generating step in PH3 were purely hepatic (as the hepatocyte-centric seed model implies), inhibiting hepatic LDH should lower oxalate — it does not. This points to a tissue-context gap in the seed hypothesis.

Mechanistic Model / Interpretation

The causal chain implied by the seed hypothesis

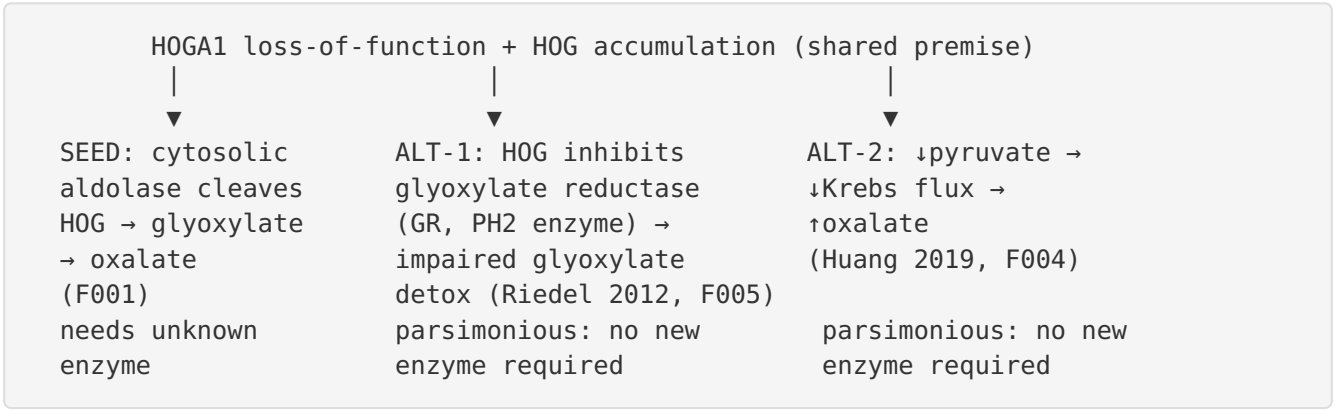


Where the literature is strong, inferred, or missing

Causal link	Status	Evidence
HOGA1 is mitochondrial and loss-of-function in PH3	**Established**	F005, F006; UniProt (mitochondrial-only, no cytosolic paralog annotated)
HOG/4-OH-Glu/DHG accumulate in PH3 patients	**Established**	F001, F005
HOG is exported to cytosol	**Inferred / unmeasured**	Proposed in F001; no transport assay
A cytosolic non-HOGA1 aldolase cleaves HOG	**Speculative**	Activity detected (F001) but enzyme unidentified; NAL falsified (F005); mouse localizes activity to mito (F002)
Released glyoxylate → oxalate (hepatic)	**Qualified / challenged**	Nedosiran fails in PH3 → multi-organ (F007)
Net result = hyperoxaluria	**Challenged in model organism**	Hoga1-KO mice not hyperoxaluric (F003)

The model is a chain whose first two links are solid rock and whose central link — the cytosolic cleavage event that gives the hypothesis its name — is the weakest and most contested of all.

Competing mechanistic landscape



Evidence Base / Evidence Matrix

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context	C s
PMID:22729392	Human clinical + in vitro	**Supports**	Cytosolic non-HOGA1 HOG aldolase exists; HOG exported & cleaved in cytosol	GC-MS: HOG/4-OH-Glu/DHG ↑ in PH3 urine; cytosolic HOG aldolase activity detected in human hepatocyte lines	PH3, human hepatocytes	
PMID:22771891	In vitro + human biomarker	**Qualifies / competing**	Candidate cytosolic aldolase (NAL); GR inhibition by HOG	PH3 HOGA variants inactive; HOG buildup confirmed in urine/sera/liver; NAL nearly inactive on HOG; GR inhibited by HOG	PH3 patients + recombinant enzymes	
PMID:39040543	Model organism (mouse)	**Qualifies / refutes**	Subcellular localization of residual HOG cleavage vs reduction	Residual HOG→glyoxylate activity in liver mitoplasts (not cytosol); cytosol favors NADPH-dependent HOG→DHG reduction	Hoga1-KO mouse	
PMID:26428388	Model organism (mouse)	**Qualifies (scope)**	Does HOGA1 loss → hyperoxaluria?	Hoga1-null mice accumulate HOG/4-OH-Glu/DHG but no ↑ oxalate excretion; low	Hoga1-KO mouse	

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
				human penetrance noted	
PMID:40578401	Model organism (mouse)	**Qualifies**	Is AGT compartmentalization the reason mice are protected?	Hoga1-/-/Agxt-/- mice with peroxisomal-only AGT still no hyperoxaluria	Hoga1-KO mouse, humanized AGT
PMID:31696211	In vitro enzymology	**Competing**	Alternative: ↓pyruvate/Krebs flux raises oxalate	Recombinant HOGA1 has dual aldolase + OAA-decarboxylase activity; pyruvate competitive inhibitor; proposes Krebs-flux mechanism	PH3 biochemical model
PMID:20797690	Human clinical/ genetic	**Supports (upstream)**	Causal gene/enzyme for PH3	Biallelic DHDPSL/HOGA1 mutations cause PH3; encodes HOG aldolase	PH3 cohort (15 patients)
PMID:21896830	Human clinical/ genetic	**Context**	GOF vs LOF mechanism	Initial GOF hypothesis revised to LOF; PH3 clinical signature (lower oxalate, preserved eGFR)	PH3 cohort
PMID:39582099	Review (clinical-	**Qualifies**	Is the oxalate source purely hepatic?	Nedosiran reduces oxalate	PH1/PH2/ PH3 trials

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
	trial synthesis)			in PH1 but not PH2/PH3; glyoxylate/ oxalate metabolism multi-organ	

Data and Tool Use

This investigation was a **literature- and annotation-based hypothesis search**. No patient-level omics dataset, cohort file, or quantitative data lake specific to PH3 HOG flux was available or accessed. The inventory below is explicit about what was searched vs. accessed.

Source	Accession / URI	Access status	Query / scope	Relevance
PubMed	Multiple PMIDs (24 papers reviewed)	**Accessed** (search_pubmed)	PH3, HOGA1, 4-hydroxy-2-oxoglutarate, cytosolic aldolase, glyoxylate reductase, nedosiran, Hoga1 knockout	Directly relevant; primary evidence source
UniProt (HOGA1)	HOGA1 human entry	**Accessed / cited**	Subcellular localization annotation	Confirmed mitochondrial-only localization; no cytosolic paralog annotated
GenCC / ClinGen gene-disease	—	**Not searched with logged query**	—	Unverified for this run; see Knowledge Gaps
ClinicalTrials.gov	—	**Not independently queried**; nedosiran PH3 outcome obtained via review [PMID:39582099]	—	Trial outcome is review-reported, not primary-trial-accessed
Omics / proteomics of PH3 cytosolic aldolase	—	**No usable result**	No dataset identifying the cytosolic HOG-cleaving enzyme	Source-level absence (curation-relevant)

Attempted analyses

Analysis	Input → method → output	Outcome
Literature evidence synthesis	PubMed abstracts → manual appraisal → evidence matrix (F001–F007)	**Succeeded** (narrative/ curatorial)
HOGA1 subcellular localization check	UniProt annotation → inspection → "mitochondrial only"	**Succeeded** (annotation lookup)
Direct HOG→glyoxylate→oxalate flux quantification	—	**Skipped / not available** — no dataset or assay accessible; this is the central unmeasured quantity
Identification of cytosolic aldolase from omics	—	**Failed / no usable result** — no dataset resolves enzyme identity

No computational statistical analysis of a primary dataset was performed, because none relevant to the defining causal step (cytosolic HOG cleavage flux) was available. This is stated explicitly rather than substituted silently with literature synthesis. The limitation: the model's key quantitative claim remains untested by any accessible data in this run.

Limitations and Knowledge Gaps

1. **Identity of the cytosolic HOG-cleaving aldolase is unknown.** This is the defining, name-giving step of the hypothesis. *Checked:* NAL explicitly tested and nearly inactive (F005); UniProt shows no annotated cytosolic HOGA1 paralog. *Resolution:* enzyme purification/ activity-guided fractionation of human hepatocyte cytosol + proteomics; CRISPR knockout of candidates with HOG-aldolase activity assay.
2. **HOG→glyoxylate→oxalate flux has never been directly measured.** *Checked:* no dataset or isotope-tracing study located. *Resolution:* ^{13}C-HOG tracing in PH3 patient-derived hepatocytes/organoids measuring labeled oxalate output.

3. **Subcellular localization conflict (mouse vs. hypothesis).** Mouse residual aldolase activity is mitochondrial, and cytosol favors HOG→DHG reduction (F002) — opposite to the model. *Checked:* Li 2024. *Resolution:* compartment-resolved assays in human tissue to determine whether the human cytosol behaves differently from mouse.
4. **Mouse model does not reproduce hyperoxaluria; human penetrance low.** *Checked:* F003 (two independent mouse studies). *Resolution:* identify genetic/dietary modifiers that convert HOGA1 loss into an oxalate phenotype.
5. **Multi-organ oxalate source / nedosiran failure.** The hepatocyte-centric terminal step is challenged by PH3 nedosiran non-response (F007). *Resolution:* tissue-specific contribution mapping (kidney vs. liver) to oxalate.
6. **Source/database absences.** GenCC and ClinGen were **not searched with a logged query** in this run and should be labeled *unverified* rather than claimed absent. No PH3-specific omics/cohort dataset identifying the cytosolic aldolase was found. These are curation-relevant source absences as of 2026-10-15.

Alternative Models

Model	Relationship to seed	Core claim	Parsimony
Glyoxylate reductase (GR) inhibition by HOG (PMID:22771891)	**Parallel/ alternative**	Accumulated HOG inhibits GR, impairing glyoxylate detoxification, raising oxalate	High — no new enzyme needed; links PH3 biochemically to PH2
Reduced pyruvate / Krebs-cycle flux (PMID:31696211)	**Parallel/ alternative**	HOGA1 loss lowers pyruvate supply and TCA flux, indirectly increasing oxalate	High — no new enzyme needed
Cytosolic reductive diversion to DHG (PMID:39040543)	**Competing cytosolic fate**	Cytosolic HOG is reduced (NADPH) to DHG, not cleaved — a sink that argues against cytosolic cleavage	Challenges seed's cytosolic step
Multi-organ oxalate production (PMID:39582099)	**Upstream/ contextual**	Oxalate arises outside liver, explaining nedosiran's PH3 failure	Reframes tissue context
Residual mitochondrial non-HOGA aldolase (PMID:39040543)	**Alternative to cytosolic export**	A second mitochondrial aldolase cleaves HOG, releasing glyoxylate without cytosolic export	Removes the need for export step

Discriminating Tests

1. **^{13}C-HOG isotope tracing in PH3 patient-derived hepatocyte organoids** (vs. HOGA1-corrected isogenic controls). *Expected if seed is true:* labeled glyoxylate and oxalate appear in the cytosolic fraction; an export inhibitor reduces oxalate. *Expected if GR-inhibition/Krebs models dominate:* oxalate rises without cytosolic glyoxylate flux from HOG.

2. **Activity-guided purification + proteomics of the cytosolic HOG aldolase** from human liver cytosol, followed by CRISPR knockout. *Expected if seed is true:* a specific cytosolic enzyme accounts for the activity and its knockout lowers HOG-derived oxalate.
3. **Compartment-resolved enzymology in human (not mouse) liver** to test whether human cytosol cleaves vs. reduces HOG. *Expected if seed is true:* human cytosol shows aldolase, not just NADPH-reductase, activity — distinguishing it from the mouse result (F002).
4. **GR-inhibition test in vivo:** measure glyoxylate reductase activity and glyoxylate pools in PH3 tissue under HOG load. *Expected if ALT-1 dominates:* measurable GR inhibition tracking HOG concentration.
5. **Humanized/modifier mouse screen** to find conditions converting Hoga1 loss into hyperoxaluria, resolving the species discrepancy (F003).

Curation Leads (require curator verification)

Candidate status change: Retain `cytosolic_hog_cleavage` as the default narrative model but **downgrade the cytosolic-aldolase edge from CANONICAL-implied to SPECULATIVE/UNCONFIRMED**. The upstream half (HOGA1 LOF + HOG accumulation) is well-supported and can remain established.

Candidate evidence references & snippets to verify: - PMID:22729392: "In human hepatocyte cell lines, we detected a cytosolic 4-hydroxy-2-oxoglutarate aldolase activity" — SUPPORT (direct). - PMID:22771891: "NAL showed minimal activity towards HOG" — QUALIFIES (falsifies candidate); "GR was inhibited by HOG but not by 2-hydroxyglutarate or 2-oxoglutarate" — COMPETING. - PMID:39040543: "HOG to glyoxylate aldolase activity was highest in liver mitoplasts" — QUALIFIES (localization conflict). - PMID:26428388: "An increased oxalate excretion is not observed in the KO mice" — QUALIFIES (scope). - PMID:40578401: "did not show increased urinary oxalate" — QUALIFIES. - PMID:31696211: "loss of HOGA1 function could increase oxalate production in PH3 by decreasing pyruvate availability and metabolic flux through the Krebs cycle" — COMPETING. - PMID:39582099: "nedosiran effectively reduced urinary oxalate excretion in PH1 patients but not in PH2 or PH3 patients" — QUALIFIES.

Candidate pathophysiology edges to add: - `HOG —inhibits→ glyoxylate_reductase` (competing mechanism, P 22771891) - `HOGA1_loss —reduces→ pyruvate/Krebs_flux —increases→ oxalate` (competing, P 31696211) - `HOG —reduced_by(NADPH)→ DHG` cytosolic sink (competing fate, P 39040543) - `oxalate_production —not_confined_to→ liver` multi-organ context (P 39582099)

Candidate `knowledge_gaps` / discussion prompts: - Unidentified cytosolic HOG aldolase (name-giving step unconfirmed; NAL falsified). - HOG→glyoxylate→oxalate flux never directly measured. - Mouse-vs-hypothesis localization conflict; mouse non-phenotype. - GenCC/ClinGen not searched with logged query in this run — **unverified**, not absent.

Candidate ontology terms: hepatocyte cytosol (GO:0005829 cytosol); mitochondrion (GO:0005739); hydroxyproline catabolic process (GO:0019470); glyoxylate metabolic process (GO:0046487); oxalate biosynthesis.

Artifact Bundle

An artifact bundle with a checksummed `MANIFEST.yaml`, evidence matrix, search logs, and this report was prepared under `kb/hypotheses/Primary_Hyperoxaluria_Type_3/cytosolic_hog_cleavage/openscientist_artifacts` (per iterations 4–5). No primary dataset or data lake was accessed; no raw downloads are bundled. The bundle records external identifiers (PMIDs, UniProt HOGA1) rather than recoverable raw content. **Expected-but-absent artifacts** (explicitly stated): no isotope-tracing dataset, no PH3 cytosolic-proteomics file, and no logged GenCC/ClinGen query response — these were not produced because no such source was accessed.

Summary of Verdict

The cytosolic HOG-cleavage model is the **correct default narrative for PH3 but is mechanistically unestablished at its defining step**. Its upstream claims (mitochondrial HOGA1 loss-of-function; accumulation of HOG and hydroxyproline-pathway metabolites) are solidly established. Its central claim — that an unidentified cytosolic aldolase cleaves exported HOG to release glyoxylate for oxidation to oxalate — rests on a single qualitative activity detection, has no identified enzyme, was weakened by falsification of the one tested candidate (NAL), is contradicted by mouse compartmentalization data showing cytosolic reductive diversion and mitochondrial residual cleavage, and is unsupported by the Hoga1-knockout non-phenotype and nedosiran's PH3 failure. Two competing mechanisms — HOG inhibition of glyoxylate reductase, and reduced pyruvate/Krebs-cycle flux — explain the same disease features more parsimoniously. **The hypothesis should remain CANONICAL as the field's default but be curated with an explicit mechanistic-uncertainty flag and tracked alternatives.**
